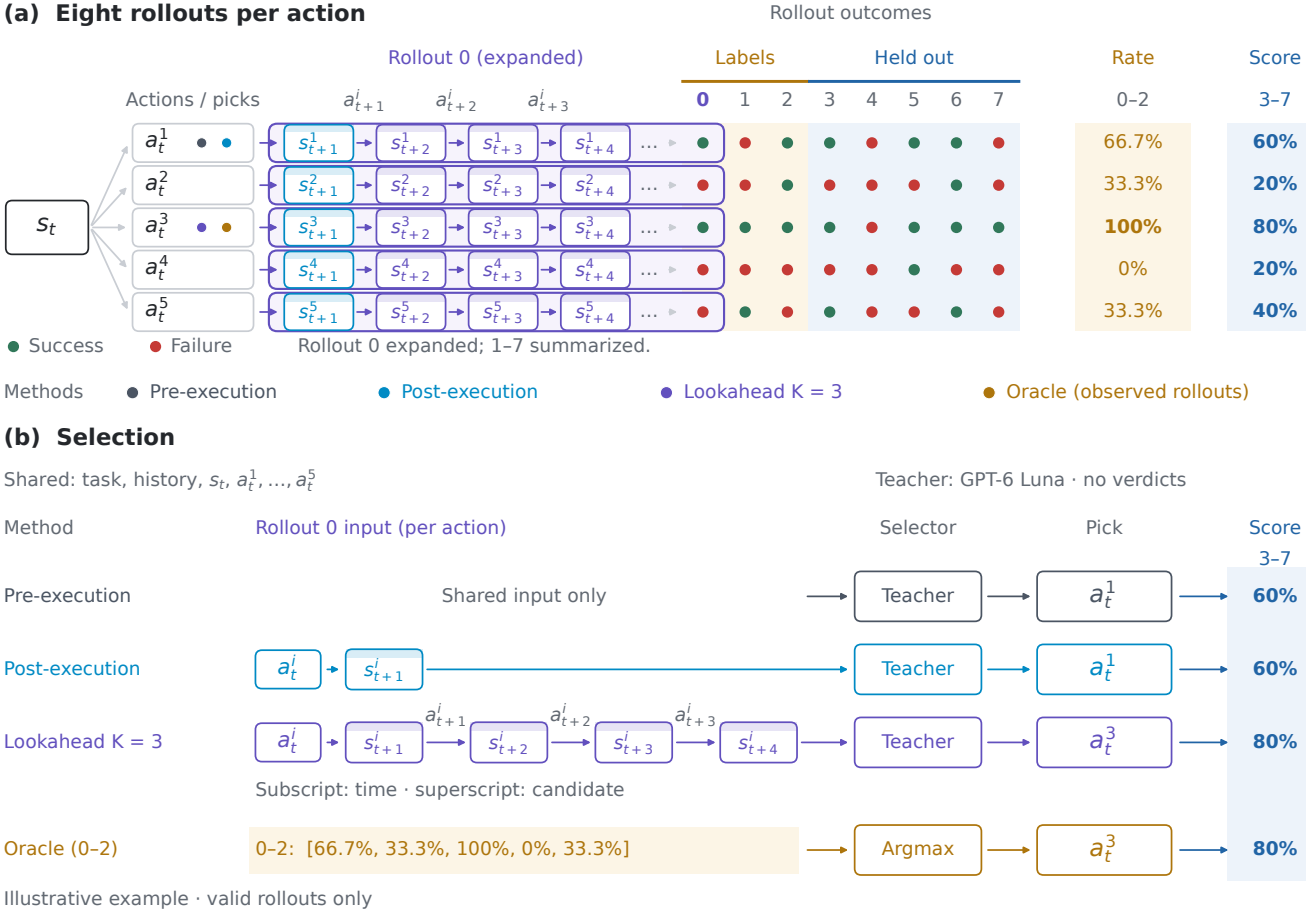
(a) Eight rollouts per action
Rollout outcomes
Rollout 0 (expanded)
Labels
Held out
Rate
Score
Actions / picks
0 1 2 3 4 5 6 7
0–2
3–7
66.7%
60%
33.3%
20%
100%
80%
0%
20%
33.3%
40%
Success
Failure
Rollout 0 expanded; 1–7 summarized.
Methods
Pre-execution
Post-execution
Lookahead K = 3
Oracle (observed rollouts)
(b) Selection
Shared: task, history, $s_t$, $a_t^1$, …, $a_t^5$
Teacher: GPT-6 Luna · no verdicts
Method
Rollout 0 input (per action)
Selector
Pick
Score
3–7
Pre-execution
Shared input only
Teacher
60%
Post-execution
Teacher
60%
Lookahead K = 3
Teacher
80%
Subscript: time · superscript: candidate
Oracle (0–2)
0–2: [66.7%, 33.3%, 100%, 0%, 33.3%]
Argmax
80%
Illustrative example · valid rollouts only